

Fiction Writing Inspired Injection Attacks in LLMs

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Literature Review: Open Problems and Literature Gaps

[Che+25]

Paper summary:

- Attacks rely on LLM ability to follow instructions + inability to separate prompts from user data
- Structured queries: general approach to solve the problem. Components:
 - (1) Secure front-end
 - (2) Specially trained LLM -> *structured instruction tuned LLM*
- Code available in the paper! :)
- Training process:
 - Augment standard instruction tuning dataset so that examples have instructions in the **data** part, train the LLM to ignore these instructions as they are not in the **prompt** part.
 - Train new LLM entirely from scratch - infeasible. Structured instruction tuning can be applied to **base LLMs**, so it's not from scratch, it's a new form of instruction tuning.
- Special separators, reserved tokens, front-end filters out these tokens from user prompt to avoid faking them
- Initialize embedding for the special token before tuning is important to keep model performance

Suggested future work:

- StruQ decreases ASR of manual attacks to 0% but some optimized attacks (TAP, GCG) still have high ASR

- StruQ is trained on task-agnostic manual injections, doesn't fully generalize to task-specific injections
 - Possible solution: craft task-dependent injections dataset with LLMs or human crowdsourcing
- StruQ protects programmatic applications that use API/libraries, but web chat-bot users wouldn't like manually marking their inputs as prompt x data constantly
- Prompt injection focus: StruQ doesn't prevent jailbreak, data extraction or other attacks, just prompt injection

[Gon+25]

Summary:

- Automated black-box jailbreaking framework inspired by SWE fuzzy testing concepts
- Proprietary APIs e.g. Gemini, GPT
- Code available
- Differential:
 - This framework can start the jailbreaking process from an empty seed, no need for starter templates
 - Aims at producing natural, semantically coherent prompts, to bypass detection
 - Aims to shorten prompts -> attacks get cheaper because proprietary APIs charge on a token basis
- White-box jailbreaking x black-box jailbreaking distinction
- White-box well-known attack: GCG
- Black-box subdivides into handcrafted and automatic
- Jailbreaking defense types
 - Input modification
 - Prompt engineering
 - Output filtering
- Defenses Papillon breaks
 - perplexity
 - SmoothLLM
 - LlamaGuard

Suggested future work:

- Add more mutators to the framework for prompt evolution
- Evaluate different prompt compression methods
- Develop defenses
- Test Papillon on non-English LLM

[PKA25]

Summary:

- Fingerprinting LLM = figuring out what LLM is being used by an app -> analogous to OS fingerprinting
- How it works
 - Send carefully crafted queries to the app
 - Analyze responses
 - Identify specific LLM version in use
- 42 different LLM versions 95% accuracy just 8 interactions
- Why is it an attack?
 - Fingerprinting enables other attack surfaces, it's related to the **reconnaissance** stage of an attack
 - Knowing the model enables trying target adversarial prompts, vulnerabilities specific to that model etc
- Tool for AI red teams
- Code available
- Queries that trigger unique response patterns that might identify what LLM is responding
 - Meta information (“what is your version?”)
 - Banner grabbing (“what is your name?”)
 - Malformed
 - Alignment / harmful / controversial
 - Nonsensical, semantically broken
- Not necessarily the model answers truthfully to a banner grabbing, but it might have a *unique way to deflect the question*
- Collect traces from the queries & responses
- Use an ML inference model (handle diverse/variable responses)
- Open-set x closed-set
 - Closed-set: classifier
 - Open-set: siamese-like, contrastive learning (“is this LLM the same as this other LLM?”)
- Prompt configuration universe -> ways LLMs get customized and become unique for a certain app
 - Hyperparameters (temperature, frequency penalty)
 - System prompt
 - Framework (RAG, COT, other...)

- Some defenses do reduce the accuracy of fingerprinting, but they have disadvantages, such as impacting model utility
- Findings suggest fingerprinting is an inevitable consequence of unique behavior exhibited by different models, unlike OS fingerprinting, which is avoidable with some measures

Suggested future work:

- GlitchTokens
- Automated query generation with optimization heuristics
- Additional inference - not only model + version, but also
 - tool call list
 - RAG / other internal framework being used
 - hyperparameters

[Bis+26a]

to-do - lido no tablet

[Bis+26b]

to-do - lido no tablet

[IBM25]

A IBM web page explaining the basics of prompt injection - what it is, how it generally works, risks etc. Not too technical, not a research paper with experimental results.

[Alb+25]

to-do - tem detalhes nos slides que apresentei no Horus

[Ber24],[Il 24]

Video and news article about someone winning \$50k by making AI hallucinate on the Freysa competition.

[Hei+24]

não cheguei a ler, tava na lista antiga do google docs

[Kie+21]

acho que eu li sim, deve estar no Drive ou no tablet, só não lembro os detalhes agora

[Fer+22]

n li ainda

[HS23]

n li ainda

[Car+23]

li sim, é o paper do Anderson e do João, é um geralzão de tudo que tem rolado de realidade sintética

[Far16]

n li, é o livro de forense digital que fala das técnicas pré-LLM, Anderson tinha recomendado

[Guo+25b]

to-do - li sim, tem detalhes em slide e no google docs, dá pra pegar

Some papers about KG + LLM, didn't read

[Ibr+26], [Kho+24], [Dai+25], [Kau+24], [Pan+24a]

[Ng17]

Curso de Deep Learning, relevante, gastei um tempão fazendo pra me especializar

Papers about anxiety in LLMs

[Cod+24] [Ben+25]

Milestone papers about LLMs

These are really cool to see the historical trajectory, taken from this repository with several LLM resources and papers: [Han24]

- [Vas+23] - Attention Is All You Need
- [RN18] - Improving Language Understanding by Generative Pre-Training
- [Dev+19] - BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
- [Rad+19] - Language Models are Unsupervised Multitask Learners
- [Sho+20] - Megatron-LM: Training Multi-Billion Parameter Language Models Using Model Parallelism
- [Raf+20] - Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer
- [Raj+20] - ZeRO: Memory Optimizations Toward Training Trillion Parameter Models
- [Kap+20] - Scaling Laws for Neural Language Models
- [Bro+20] - Language Models are Few-Shot Learners
- [FZS22] - Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity
- [Che+21] - Evaluating Large Language Models Trained on Code
- [Bom+22] - On the Opportunities and Risks of Foundation Models
- [Wei+22b] - Finetuned Language Models Are Zero-Shot Learners
- [San+22] - Multitask Prompted Training Enables Zero-Shot Task Generalization
- [Du+22] - GLaM: Efficient Scaling of Language Models with Mixture-of-Experts
- [Nak+21] - WebGPT: Browser-assisted question-answering with human feedback
- [Bor+22] - Improving language models by retrieving from trillions of tokens
- [Rae+22] - Scaling Language Models: Methods, Analysis & Insights from Training Gopher
- [Wei+23] - Chain-of-Thought Prompting Elicits Reasoning in Large Language Models
- [Tho+22] - LaMDA: Language Models for Dialog Applications
- [Lew+22] - Solving Quantitative Reasoning Problems with Language Models
- [Smi+22] - Using DeepSpeed and Megatron to Train Megatron-Turing NLG 530B, A Large-Scale Generative Language Model
- [Ouy+22] - Training language models to follow instructions with human feedback
- [Cho+22] - PaLM: Scaling Language Modeling with Pathways
- [Hof+22] - Training Compute-Optimal Large Language Models
- [Zha+22] - OPT: Open Pre-trained Transformer Language Models
- [Tay+23] - UL2: Unifying Language Learning Paradigms
- [Wei+22a] - Emergent Abilities of Large Language Models
- [Sri+23] - Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models
- [Hao+22] - Language Models are General-Purpose Interfaces
- [Gla+22] - Improving alignment of dialogue agents via targeted human judgements
- [Chu+22] - Scaling Instruction-Finetuned Language Models

- [Zen+23] - GLM-130B: An Open Bilingual Pre-trained Model
- [Lia+23] - Holistic Evaluation of Language Models
- [Wor+23] - BLOOM: A 176B-Parameter Open-Access Multilingual Language Model
- [Tay+22] - Galactica: A Large Language Model for Science
- [Iye+23] - OPT-IML: Scaling Language Model Instruction Meta Learning through the Lens of Generalization
- [Lon+23] - The Flan Collection: Designing Data and Methods for Effective Instruction Tuning
- [Tou+23b] - LLaMA: Open and Efficient Foundation Language Models
- [Hua+23] - Language Is Not All You Need: Aligning Perception with Language Models
- [Orv+23] - Resurrecting Recurrent Neural Networks for Long Sequences
- [Dri+23] - PaLM-E: An Embodied Multimodal Language Model
- [Ope+24] - GPT-4 Technical Report
- [Liu+23] - Visual Instruction Tuning
- [Bid+23] - Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling
- [Sun+23] - Principle-Driven Self-Alignment of Language Models from Scratch with Minimal Human Supervision
- [Ani+23] - PaLM 2 Technical Report
- [Pen+23] - RWKV: Reinventing RNNs for the Transformer Era
- [Raf+24] - Direct Preference Optimization: Your Language Model is Secretly a Reward Model
- [Yao+23] - Tree of Thoughts: Deliberate Problem Solving with Large Language Models
- [Tou+23a] - Llama 2: Open Foundation and Fine-Tuned Chat Models
- [Jia+23] - Mistral 7B
- [GD24] - Mamba: Linear-Time Sequence Modeling with Selective State Spaces
- [Dee+24] - DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model
- [Gro+24] - OLMo: Accelerating the Science of Language Models
- [DG24] - Transformers are SSMS: Generalized Models and Efficient Algorithms Through Structured State Space Duality
- [Gra+24] - The Llama 3 Herd of Models
- [Pen+24] - The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale
- [Mue+25] - OLMoE: Open Mixture-of-Experts Language Models
- [Qwe+25] - Qwen2.5 Technical Report
- [Dee+25] - DeepSeek-V3 Technical Report
- [Guo+25a] - DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning

Google Research

- [Zho+25] - InstructPipe: Generating Visual Blocks Pipelines with Human Instructions and LLMs

- [Mou+25] - Gemini & Physical World: Large Language Models Can Estimate the Intensity of Earthquake Shaking from Multi-Modal Social Media Posts
- [Wu+25] - SSDTrain: Faster Large Language Model Training Using SSD-Based Activation Offloading
- [Fri+25] - Triaging mammography with artificial intelligence: an implementation study
- [Lim+25] - Mufu: Multilingual Fused Learning for Low-Resource Translation with LLM
- [Wan+25a] - TOKENFORMER: Rethinking Transformers Scaling with Tokenized Model Parameters
- [MB24] - Generative Ghosts: Anticipating Benefits and Risks of AI Afterlives
- [Lee+25] - Gemini Embedding: Generalizable Embeddings from Gemini
- [Man+25] - TIPS: Text-Image Pretraining with Spatial awareness
- [Wan+25b] - Scaling Pre-training to One Hundred Billion Data for Vision Language Models
- [Coo+25] - Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy and Research
- [Sch+25] - Mastering Board Games by External and Internal Planning with Language Models
- [DAm+25] - Achieving Human Level Competitive Robot Table Tennis
- [HQC24] - An Introduction to Universal Artificial Intelligence
- [Pan+24b] - Teach LLMs to Phish: Stealing Private Information from Language Models
- [Whi+24] - Learning 3D Particle-based Simulators from RGB-D Videos
- [Son+24] - Position: Leverage Foundational Models for Black-Box Optimization

Microsoft Research

- [Vor+25] - Fostering appropriate reliance on GenAI: Lessons learned from early research
- [Flo+25] - The Impact of Generative AI Coding Assistants on Developers Who Are Visually Impaired
- [Als+25] - A Unifying Framework for Representation Learning

OpenAI

- [Mis+25] - SWE-Lancer: Can Frontier LLMs Earn \$1 Million from Real-World Freelance Software Engineering?
- [Bak+22] - Video PreTraining (VPT): Learning to Act by Watching Unlabeled Online Videos
- [Rad+21] - Learning Transferable Visual Models From Natural Language Supervision (CLIP)
- [Ope21] - DALL · E: Creating Images from Text
- [Ope18] - Learning Montezuma's Revenge from a single demonstration
- [Ope22] - Lessons learned on language model safety and misuse

- [Goh+21] - Multimodal Neurons in Artificial Neural Networks

Synthetic Media Detection

- [CDY20] - FakeCatcher: Detection of Synthetic Portrait Videos using Biological Signals
- [Wan23] - Synthetic Realities in the Digital Age: Navigating the Opportunities and Challenges of AI-Generated Content
- [Ye+25] - LOKI: A Comprehensive Synthetic Data Detection Benchmark using Large Multimodal Models
- [Zob+21] - DeepFakes: Detecting Forged and Synthetic Media Content Using Machine Learning
- [Cor+23] - On The Detection of Synthetic Images Generated by Diffusion Models
- [BL24] - CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images
- [Lau+24] - The Cat and Mouse Game: The Ongoing Arms Race Between Diffusion Models and Detection Methods

Ilya Sutskever's Recommended Reading List

Papers and resources from [dzy23], originally shared by Ilya Sutskever with John Carmack in 2020.

- [Rus+18] - The Annotated Transformer
- [Aar11] - The First Law of Complexodynamics
- [Kar15] - The Unreasonable Effectiveness of Recurrent Neural Networks
- [Ola15] - Understanding LSTM Networks
- [ZSV14] - Recurrent Neural Network Regularization
- [HC93] - Keeping the Neural Networks Simple by Minimizing the Description Length of the Weights
- [VFJ15] - Pointer Networks
- [KSH12] - ImageNet Classification with Deep Convolutional Neural Networks
- [VBK16] - Order Matters: Sequence to Sequence for Sets
- [Hua+19] - GPipe: Easy Scaling with Micro-Batch Pipeline Parallelism
- [He+15] - Deep Residual Learning for Image Recognition
- [YK16] - Multi-Scale Context Aggregation by Dilated Convolutions
- [Gil+17] - Neural Message Passing for Quantum Chemistry
- [BCB16] - Neural Machine Translation by Jointly Learning to Align and Translate
- [He+16] - Identity Mappings in Deep Residual Networks
- [San+17] - A Simple Neural Network Module for Relational Reasoning
- [Che+17] - Variational Lossy Autoencoder
- [San+18] - Relational Recurrent Neural Networks

- [ACO14] - Quantifying the Rise and Fall of Complexity in Closed Systems: The Coffee Automaton
- [GWD14] - Neural Turing Machines
- [Amo+16] - Deep Speech 2: End-to-End Speech Recognition in English and Mandarin
- [Grü04] - A Tutorial Introduction to the Minimum Description Length Principle
- [Leg08] - Machine Super Intelligence (PhD Thesis)
- [SUV17] - Kolmogorov Complexity and Algorithmic Randomness
- [Li+16] - CS231n: Convolutional Neural Networks for Visual Recognition

Misc, to be organized

- [Kum+23] - Language Generation Models Can Cause Harm: So What Can We Do About It? An Actionable Survey
- [Agr+24] - Can Knowledge Graphs Reduce Hallucinations in LLMs? : A Survey
- [FZF23] - Knowledge Solver: Teaching LLMs to Search for Domain Knowledge from Knowledge Graphs
- [Lu+24] - The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery
- [CPR24] - Durably reducing conspiracy beliefs through dialogues with AI
- [Nak25] - Scalable-Softmax Is Superior for Attention
- [Fen+24] - Were RNNs All We Needed?
- [Zha+25] - From Thought to Action: How a Hierarchy of Neural Dynamics Supports Language Production
- [Lév+25] - Brain-to-Text Decoding: A Non-invasive Approach via Typing
- [Li+25] - Loki: An Open-Source Tool for Fact Verification

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